

Hyperspectral Object Detection Challenge 2026

Public Resources -> Controlled Experiments -> Claude Implementation Plan

Research brief for Kaggle competition engineering
Verified / curated on 26 September 2026

[Competition page](#)

Core objective: build one robust hyperspectral-aware detector that generalizes to the hidden/ranking data. Public leaderboard score is secondary to leakage-safe CV, spectral robustness, and single-checkpoint rule compliance.

1. Executive Summary

- The challenge uses 16-band visible hyperspectral imagery spanning approximately 460-600 nm and requires bounding-box localization plus 18-class prediction.
- The primary metric is mAP@[0.5:0.95], so localization quality at high IoU thresholds matters as much as class accuracy.
- The strongest public analogue found is HOD3K/S2ADet: HOD3K uses 16 bands over 470-620 nm, unusually close to the challenge spectral regime.
- The final solution must be generated by one trained model / one checkpoint. Same-model TTA and multi-scale inference are allowed; multi-model fusion is not eligible under the organizer clarification.
- Self-training on unlabeled test images is permitted; ranking images are inference-only and cannot update model parameters or statistics.
- The research priority is not to test 100 models. It is to use 100+ resources to identify a small set of high-information ablations that isolate where gains come from: input representation, spectral modeling, localization, pretraining, pseudo-labels, and inference.
- Recommended development path: clean YOLO/DETR baseline -> spectral input ablation -> S2ADet-style dual stream -> spectral derivative branch -> HSI foundation model backbone -> small-object/localization tuning -> conservative pseudo-labeling -> same-checkpoint TTA.

2. Verified Competition Facts and Hard Constraints

Item	Current verified interpretation	Source
Input	16-band visible-light hyperspectral imagery	Competition overview
Spectral range	Approximately 460-600 nm	Competition overview
Targets	18 object categories, including difficult real-vs-counterfeit pairs	Competition overview
Primary metric	mAP@[0.5:0.95]	Competition evaluation
Secondary metric	mAP@0.5	Competition evaluation
Final evaluation	Test set and ranking set are weighted equally in final score	Competition overview
Final model rule	One trained model / one checkpoint	Organizer clarification
TTA	Same-model flip/scale/crop TTA is allowed	Organizer clarification
Public pretrained weights	Allowed when declared with model/source/license	Organizer clarification
Unlabeled test self-training	Permitted; document teacher, threshold, filtering, iterations	Organizer clarification
Ranking data	Inference only; no training, pseudo-labeling, BN update, entropy minimization or TTA that changes model state	Organizer clarification
External labelled data	Not allowed for training except declared public pretrained weights	Organizer clarification

Competition: <https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026>

Single-model / pretrained clarification: <https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/discussion/727863>

Pseudo-label / ranking-set clarification: <https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/discussion/739853>

Leaderboard: <https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/leaderboard>

3. What the 100+ Resources Suggest

The literature and public code point to six reusable ideas. The competition-specific hypothesis is that material-sensitive classification needs more than an RGB-style spatial detector, while the mAP50-95 metric still demands precise spatial localization.

A. Spectral-spatial decoupling: S2ADet explicitly separates hyperspectral information into complementary spectral and spatial representations before detection. This is the clearest direct architecture analogue.

B. Native cube modeling: SpecDETR and modern HSI models show that spatial-spectral features can be learned directly from HSI cubes rather than reducing the input to pseudo-RGB only.

C. Wavelength/channel-adaptive pretraining: HyperVision, HyperFree, HyperSL and DOFA-style models motivate flexible channel/wavelength embeddings rather than forcing 16 bands into a fixed RGB stem.

D. Spectral derivative cues: Material differences can appear as local spectral-shape changes. A first-derivative branch is cheap, interpretable, and high-information to test.

E. Localization-specific optimization: Because AP is averaged through IoU 0.95, high-resolution features, P2/FPN choices, assignment and box loss need independent evaluation.

F. Semi-supervision with a legal boundary: The test set can be pseudo-labeled, but ranking images are held out. Pseudo-labeling must therefore prove robustness under train-only CV and stress splits before adoption.

4. Highest-Value Resources -> Exact Experiments

Resource	Type	URL	Exact transferable value	Experiment	Rule note
S2ADet / HOD3K	GitHub + TGRS	https://github.com/hexiao0275/S2ADet	Closest public analogue: 16 bands, 470-620 nm, HOD3K. Reuse HID / spectral-spatial two-stream concept.	E4: S2A-style dual-stream detector	Research inspiration; external labels not for training.
SpecDETR	GitHub + ISPRS P&RS	https://github.com/ZhaoxuLi123/SpecDETR	Native HSI point/tiny-object DETR; provides configs for mainstream detectors on HSI.	E10: SpecDETR or HSI-DETR benchmark	Code reusable; pretrained-weight legality must be declared/verified.
HODToolbox	GitHub	https://github.com/ZhaoxuLi123/HODToolbox	HSI detection conversion, COCO-style evaluation, visualization, HTD-to-HOD utilities.	Evaluation and error-analysis tooling	Tooling only; safe.
HyperVision	GitHub	https://github.com/lronkitty/HyperVision	Ground-based channel-adaptive HSI foundation backbone, much closer domain than satellite-only FMs.	E8: frozen/head-only then partial fine-tune backbone	Public pretrained weights likely allowed if declared; verify license.
HyperFree	GitHub + HF	https://github.com/Jingtao-Li-CVer/HyperFree	Dynamic embedding for variable band count and wavelength configuration.	E9A: wavelength-aware input embedding	Public pretrained weights likely allowed if declared; verify exact model license.
HyperSL	GitHub	https://github.com/kkweil/HyperSL	Spectral foundation model with downstream_detection.py.	E8B: HyperSL feature extractor + detection head	Weights/source/license must be documented.
HyperSIGMA	GitHub + HF	https://github.com/	Spatial-MAE +	E8C: use spectral	Likely allowed as

		WHU-Sigma/ HyperSIGMA	Spectral-MAE with spectral enhancement; strong HSI foundation-model reference.	encoder features / lightweight adaptation	declared public weights; satellite-domain mismatch risk.
SpectralGPT	GitHub + TPAMI	https://github.com/danfenghong/IEEE_TPAMI_Spectral_GPT	3D generative spectral pretraining, good reference for spectral tokenization.	E9B: spectral token/MAE-inspired stem	Research/backbone candidate; domain mismatch.
DOFA	GitHub	https://github.com/zhu-xlab/DOFA	Supports any number of channels and uses wavelength-conditioned dynamic embeddings.	E9C: explicit wavelength-conditioned 16-band stem	Strong concept; direct HSI example support should be verified before porting.
HyperMamba	GitHub + TGRS	https://github.com/chiangliu/HyperMamba	Spectral-spatial adaptive state-space model.	E7C: compact Mamba spectral encoder	Use module idea only unless checkpoint compatibility is proven.
Spa-Spe-TR	GitHub + GRSL	https://github.com/Li-Qingyun/Spa-Spe-TR	Two-branch pure transformer for separate spatial/spectral reasoning.	E7B: cross-attention/gated two-branch fusion	Research inspiration.
SAHI	GitHub	https://github.com/obss/sahi	Tiled/sliced inference toolkit for small objects.	E11: tile inference only if bbox forensics justify it	Same-checkpoint inference compatible; confirm merge method.
Ultralytics YOLO11	GitHub	https://github.com/ultralytics/ultralytics	Fast, reproducible detector baseline with COCO pretrained weights.	E0-E7 baseline family	Weights allowed if declared; software license must be respected.
MMDetection	GitHub	https://github.com/open-mmlab/mmdetection	DINO/Deformable-DETR/FCOS/Cascade framework; close to SpecDETR ecosystem.	E10 and localization ablations	Framework only.
Spectral Python	GitHub	https://github.com/spectralpython/spectral	Reliable HSI reading/processing and spectral analysis.	Data audit, spectral profiles, PCA prototypes	Tooling only.

5. Controlled Experiment Roadmap

ID	Experiment	Implementation	Question answered	Validation	Decision gate	Compute
E0	Clean detector baseline	Pseudo-RGB / deterministic 3-band selection -> YOLO11m	Establish reproducible floor and runtime	Primary CV + stress CV	Keep as reference even if weak	Low
E1	Normalization	Compare train-global per-band z-score vs robust percentile scaling	Determine whether sensor scaling is a major source of variance	Same model/seed/folds	Keep only if mean CV improves and fold std does not worsen	Low
E2	PCA vs selected bands	Fit PCA on legal train only; compare 3 PCs with pseudo-RGB	Measure information loss from 16->3 compression	Same detector	Reject if class-pair AP falls	Low
E3	Learned 16->3 adapter	1x1 Conv 16->32->3 before COCO-pretrained detector	Test whether learned spectral projection beats fixed compression	Freeze/unfreeze adapter stages	Keep if gain is stable across folds	Low-Med
E4	Native 16-channel stem	Replace first conv/stem for 16 channels; initialize by RGB weight averaging/replication	Test direct spectral access without separate branch	Exact same detector otherwise	Keep if robust CV > E3	Med
E5	S2ADet-style dual stream	Spatial stream + spectral stream + gated/concat fusion -> one head	Exploit complementary spectral/material and spatial cues	Single model / one checkpoint	High priority if E3/E4 show spectral value	Med-High

E6	Spectral derivative branch	15 adjacent-band derivatives, optionally raw+derivative dual input	Expose local spectral-shape cues for real/fake pairs	Pairwise AP + overall mAP	Keep only if pair AP and total CV improve	Low-Med
E7	Spectral attention / lightweight Mamba	SE/ECA or compact spectral Mamba before fusion	Learn informative wavelengths without hard selection	Channel ablation + CV	Reject if unstable or overfits rare classes	Med
E8	HSI foundation backbone	HyperVision first; then HyperSL/HyperSIG MA if feasible	Test transfer from HSI pretraining	Frozen head-only -> partial fine-tune	Adopt only with clear CV/stress gain	High
E9	Wavelength-aware embedding	HyperFree/DOFA-style dynamic embedding using actual band centers	Use spectral physics explicitly	Only exact metadata wavelengths	High research value; verify implementation	High
E10	SpecDETR benchmark	Port dataset to MMDetection/Spec DETR format; compare a single DETR-style model	Test if transformer localization/tiny-object handling beats YOLO family	One checkpoint only	Keep if AP75 and small-object AP improve	High
E11	Resolution / P2 / tile ablation	Increase input size; add high-res head; test SAHI only if objects are small	Improve high-IoU localization and tiny objects	AP by object size	Do not keep expensive tiling without clear benefit	Med
E12	Box loss + assignment	CloU/default vs supported alternatives; task-aligned assignment settings	Improve mAP50-95 rather than AP50 only	AP50, AP75, AP50-95	Keep only if AP75+ improves	Low-Med
E13	Spectral augmentation	Mild multiplicative scaling/noise, band dropout, contiguous masking	Improve robustness without destroying material spectra	Spectral corruption stress test	Reject aggressive transforms that harm pair AP	Low
E14	Class/rare-object handling	Repeat-factor or class-aware sampler; conservative focal/varifocal options	Address imbalance without duplicating scenes excessively	Per-class AP + fold std	Keep only if tail AP rises without head collapse	Low-Med
E15	Pseudo-label self-training	Teacher checkpoint -> high-confidence pseudo labels on allowed test images	Use legal unlabeled target-domain data	Document threshold/count/iterations	Adopt only after CV/stress safeguards	Med-High
E16	Spectral-consistency pseudo filter	Train-only class prototypes + spectral angle/cosine consistency filter	Reduce confident but spectrally implausible pseudo labels	Pseudo precision proxies + ablation	Keep only if better than confidence-only	Med
E17	Confidence/NMS calibration	Tune global or class-aware threshold and NMS on OOF only	Recover precision/recall without retraining	OOF optimization only	Never tune from repeated public-LB probes	Low
E18	Same-checkpoint TTA	Original + HFlip + selected multi-scale; merge only same checkpoint	Final legal inference gain	OOF TTA simulation	Use only if runtime acceptable and CV improves	Low-Med
E19	Champion robustness test	3 seeds + primary CV + stress CV + spectral corruption + error slices	Protect final selection from leaderboard overfit	All robustness metrics	Champion only if robust across tests	High compute

6. First 10 Experiments - Exact Order

Order	ID	Run	Why now
1	E0	Clean YOLO11m pseudo-RGB baseline	Need an auditable reference before spectral changes.
2	E1	Normalization ablation	Cheap; can invalidate every later comparison if wrong.
3	E2	PCA vs selected-band pseudo-RGB	Measures whether simple spectral compression is already sufficient.
4	E3	Learned 16->3 spectral adapter	Lowest-cost way to exploit all 16 bands while retaining COCO spatial features.
5	E4	Native 16-channel stem	Separates gain from learned projection vs truly native multi-band features.
6	E6	Spectral derivative branch	High-information test for material/real-fake discrimination.
7	E5	S2ADet-style dual stream	Only after evidence shows spectral signal matters.
8	E11	Resolution/P2/tile study	Run after bbox forensics to address AP75/localization bottleneck.
9	E8	HyperVision backbone	Most domain-aligned HSI foundation-model experiment.
10	E15+E18	Conservative pseudo-label then same-model TTA	Final-stage gains only after base model and CV are stable.

7. Private-Leaderboard-Safe Validation

- Do not use random KFold until duplicate/scene/acquisition structure is audited. Near-identical scenes across folds can produce misleadingly high mAP.
- Primary split: StratifiedGroupKFold using the strongest available grouping variable (scene/acquisition/duplicate cluster). If metadata is absent, build duplicate/scene clusters from train-only embeddings and spectral similarity.
- Stress split: hold out the most distribution-shifted scene/domain cluster; use this to reject brittle public-LB tricks.
- Fast screening: one fixed group-aware fold for low-cost ablations; every candidate must later pass the full primary CV.
- Track mAP50-95, AP50, AP75, per-class AP, object-size AP, real/fake-pair confusion, fold std and seed std.
- Submission ledger rule: if public LB rises while mean CV or stress CV worsens, flag PUBLIC-LB OVERFIT RISK and do not automatically keep the change.

8. Recommended Competitive Single-Model Pipeline

Step	Stage	Implementation note
1	Raw HSI	16-band cube; preserve wavelength order.
2	Preprocess	Train-derived robust per-band scaling; no independent test fitting.
3	Spectral inputs	Raw bands + optional adjacent first derivative.
4	Spectral encoder	Lightweight 1D/3D conv or Mamba/attention module.

5	Spatial encoder	COCO-pretrained detector backbone OR HSI foundation backbone if validated.
6	Fusion	Gated concat / 1x1 projection at early/mid FPN scales.
7	Detection	Single FPN/PAN/DETR head; 18 classes.
8	Training	Group-safe CV; geometry augmentation + conservative spectral augmentation.
9	Optional self-training	Only allowed unlabeled test images; one teacher; strict filtering; documented.
10	Inference	One checkpoint; same-model flip/multi-scale TTA if CV-positive.
11	Postprocess	OOF-tuned threshold/NMS; no multi-model fusion.
12	Submission	Strict CSV schema; test + ranking phase handling per current rules.

9. Claude / Coding-Agent Working Contract

- Never fabricate CV or leaderboard results. A result cell remains null until an actual run produces it.
- Implement experiments as diffs from a common baseline. One major variable per experiment.
- Every run saves config, git diff/hash, seed, fold, preprocessing stats, metrics, checkpoint path and prediction file.
- No external labelled dataset may be added to training unless an updated organizer clarification explicitly permits it.
- Never train/adapt on ranking images. Never re-estimate BatchNorm or other statistics on ranking data.
- Final prediction must use one trained checkpoint. Do not build fold ensembles or architecture ensembles.
- Use public pretrained weights only with model/source/license recorded in README.
- Pseudo-labeling must log teacher, image subset, threshold, filtering logic, label count and iterations.
- Any public-LB-only improvement without robust CV support is treated as inconclusive, not a success.

10. Public Resource Database - 100+ Actionable References

Tier indicates testing priority, not paper quality. "Research only" means the idea can be studied but the data/checkpoint must not be assumed legal for final training without rule verification.

#	Tier	Resource	Type	Actionable use	Direct URL
1	S	Kaggle HOD Challenge 2026 - Overview	Kaggle	Competition facts/rules	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/overview
2	S	Kaggle HOD Challenge 2026 - Data	Kaggle	Dataset files/format	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/data
3	S	Kaggle HOD Challenge 2026 - Discussion	Kaggle	Organizer clarifications	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/discussion
4	S	Kaggle HOD Challenge 2026 - Leaderboard	Kaggle	Public-LB context only	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/leaderboard
5	S	Single-model + pretrained clarification	Kaggle Discussion	Hard compliance constraint	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/discussion/727863
6	S	Pseudo-label + ranking-set clarification	Kaggle Discussion	Defines legal self-training boundary	https://www.kaggle.com/competitions/hyperspectral-object-detection-challenge-2026/discussion/739853
7	A	Public HOD26 Kaggle notebook search	Kaggle	Baseline discovery	https://www.kaggle.com/code?searchQuery=hyperspectral%20object%20detection%20challenge%202026
8	B	Hyperspectral Object Tracking Challenge 2026	Kaggle	Related sensor/task ideas	https://www.kaggle.com/competitions/hyperspectral-object-tracking-challenge-2026/overview
9	S	S2ADet official repo	GitHub	Spectral-spatial decoupling	https://github.com/hexiao0275/S2ADet
10	S	S2ADet paper	arXiv	HID + two-stream design	https://arxiv.org/abs/2306.08370
11	S	SpecDETR official repo	GitHub	Native HSI DETR/tiny objects	https://github.com/ZhaoxuLi123/SpecDETR
12	S	HODToolbox	GitHub	HSI detection eval/conversion	https://github.com/ZhaoxuLi123/HODToolbox
13	A	PBVS 2022 Semi-Supervised HSI Object Detection Challenge	CVPR Workshop	Pseudo-label/winner methodology	https://openaccess.thecvf.com/content/CVPR2022W/PBVS/html/Rangnekar_Semi-Supervised_Hyperspectral_Object_Detection_Challenge_Results_-_PBVS_2022_CVPRW_2022_paper.html
14	A	Hyperspectral OD with Spatial-Spectral Fusion and Visual Mamba	Remote Sensing	Mamba + detector fusion reference	https://www.mdpi.com/2072-4292/16/23/4482
15	A	HOD3K via S2ADet repository	Dataset	Closest 16-band public analogue; research only unless allowed	https://github.com/hexiao0275/S2ADet#dataset
16	A	SPOD benchmark via SpecDETR	Dataset	Tiny/subpixel HSI detection reference	https://github.com/ZhaoxuLi123/SpecDETR#dataset-preparation
17	S	HyperVision	GitHub	Ground-based channel-adaptive HSI backbone	https://github.com/Ironkitty/HyperVision
18	S	HyperFree	GitHub	Wavelength/channel-adaptive embedding	https://github.com/Jingtao-Li-CVer/HyperFree
19	S	HyperSL	GitHub	Spectral foundation model + downstream detection	https://github.com/kkweil/HyperSL
20	A	HyperSIGMA	GitHub	Spatial/spectral MAE + fusion	https://github.com/WHU-Sigma/HyperSIGMA

21	A	HyperSIGMA weights	Hugging Face	Public pretrained weights candidate	https://huggingface.co/WHU-Sigma/HyperSIGMA
22	A	HyperSIGMA paper	arXiv	Foundation-model architecture	https://arxiv.org/abs/2406.11519
23	A	SpectralGPT	GitHub	3D spectral token pretraining	https://github.com/danfenghong/IEEE_TPAMI_SpectralGPT
24	A	SpectralGPT paper	IEEE TPAMI	Spectral foundation model	https://doi.org/10.1109/TPAMI.2024.3362475
25	A	DOFA	GitHub	Any-channel wavelength-conditioned encoder	https://github.com/zhu-xlab/DOFA
26	A	AnySat	GitHub	Multi-sensor/modality representation	https://github.com/gastruc/AnySat
27	A	SatMAE	GitHub	Multispectral masked autoencoding	https://github.com/sustainlab-group/SatMAE
28	B	SatMAE paper	arXiv	Grouped-channel MAE reference	https://arxiv.org/abs/2207.08051
29	A	ExPLoRA	GitHub	LoRA domain-adaptation idea	https://github.com/samar-khanna/ExPLoRA
30	B	CROMA	GitHub	Contrastive + masked RS pretraining	https://github.com/antofuller/CROMA
31	B	Prithvi-EO-2.0	GitHub	EO foundation-model reference	https://github.com/NASA-IMPACT/Prithvi-EO-2.0
32	B	Prithvi-EO-2.0 300M	Hugging Face	Public model weights	https://huggingface.co/ibm-nasa-geospatial/Prithvi-EO-2.0-300M
33	B	Prithvi-EO-2.0 600M	Hugging Face	Large public model weights	https://huggingface.co/ibm-nasa-geospatial/Prithvi-EO-2.0-600M
34	B	Prithvi-EO-2.0 paper	arXiv	MAE + EO transfer	https://arxiv.org/abs/2412.02732
35	B	Clay Foundation Model	GitHub	Multi-sensor wavelength-aware EO backbone	https://github.com/Clay-foundation/model
36	B	Clay weights	Hugging Face	Public pretrained weights	https://huggingface.co/made-with-clay/Clay
37	B	TerraMind	GitHub	Multimodal EO foundation model	https://github.com/IBM/terramind
38	B	Scale-MAE	GitHub	Scale-aware remote-sensing MAE	https://github.com/bair-climate-initiative/scale-mae
39	B	RemoteCLIP	GitHub	Remote-sensing vision-language pretraining	https://github.com/ChenDelong1999/RemoteCLIP
40	B	SatlasPretrain Models	GitHub	Remote-sensing pretrained backbones/FPN	https://github.com/allenai/satlaspretrain_models
41	B	SkySense	GitHub	Multimodal remote-sensing FM	https://github.com/Jack-bo1220/SkySense
42	A	SpectralFormer	GitHub	Spectral transformer modules	https://github.com/danfenghong/IEEE_TGRS_SpectralFormer
43	A	HyperMamba	GitHub	Spectral-spatial state-space encoder	https://github.com/chiangliu/HyperMamba
44	A	HSIMamba	GitHub	Bidirectional SSM spectral feature learning	https://github.com/Judyxyang/HSImamba
45	A	Spa-Spe-TR	GitHub	Separate spatial/spectral transformer streams	https://github.com/Li-Qingyun/Spa-Spe-TR
46	B	PyFormer	GitHub	Hierarchical spatial-spectral transformer	https://github.com/mahmad000/PyFormer
47	B	3DSS-Mamba	GitHub	3D spectral-spatial Mamba idea	https://github.com/IIP-Team/3DSS-Mamba
48	B	VMamba	GitHub	Vision state-space building blocks	https://github.com/MzeroMiko/VMamba
49	B	Swin Transformer	GitHub	Hierarchical visual backbone	https://github.com/microsoft/Swin-Transformer
50	B	MAE	GitHub	Masked autoencoder adaptation reference	https://github.com/facebookresearch/mae
51	B	DINOv2	GitHub	Strong RGB self-supervised spatial backbone	https://github.com/facebookresearch/dinov2
52	B	SAM 2	GitHub	Mask-assisted ROI/spectral pooling research	https://github.com/facebookresearch/sam2
53	B	Awesome Hyperspectral Datasets	GitHub	Dataset discovery and wavelength comparisons	https://github.com/033labcodes/awesome-hyperspectral-datasets
54	B	Awesome Hyperspectral	GitHub	Tools/papers/spectral libraries	https://github.com/Nrevyw/awesome-hyperspectral

55	B	HyperspectralDatasets	GitHub	Sensor/dataset catalogue	https://github.com/antmedellin/HyperspectralDatasets
56	B	DeepHyperX	GitHub	Classical HSI deep-learning baselines	https://github.com/nshaud/DeepHyperX
57	B	HSI Classification Models collection	GitHub	Large module zoo for spectral encoders	https://github.com/Candy-CY/Hyperspectral-Image-Classification-Models
58	C	CAVE hyperspectral database	Dataset	Ground-based visible HSI reference	https://www.cs.columbia.edu/CAVE/databases/multispectral/
59	C	Harvard hyperspectral dataset	Dataset	Natural-image spectral statistics	http://vision.seas.harvard.edu/hyperspec/
60	C	ARAD-1K spectral dataset	Dataset	Visible spectral reconstruction dataset	https://github.com/boazarad/ARAD_1K
61	C	USGS Spectral Library	Spectral Library	Material spectra background/reference only	https://www.usgs.gov/labs/spec-lab/capabilities/spectral-library
62	C	ECOSTRESS Spectral Library	Spectral Library	Material spectral signatures reference	https://speclib.jpl.nasa.gov/
63	S	Ultralytics	GitHub	YOLO11 baseline/training pipeline	https://github.com/ultralytics/ultralytics
64	A	MMDetection	GitHub	DINO/DETR/FCOS/Cascade experimentation	https://github.com/open-mmlab/mmdetection
65	A	Detectron2	GitHub	Faster/Cascade/Mask R-CNN framework	https://github.com/facebookresearch/detectron2
66	A	TorchVision detection	GitHub	Faster R-CNN/RetinaNet/FCOS references	https://github.com/pytorch/vision
67	A	DINO detector	GitHub	Strong DETR-style detector	https://github.com/IDEA-Research/DINO
68	A	Deformable DETR	GitHub	Multi-scale deformable attention	https://github.com/fundamentalvision/Deformable-DETR
69	A	RT-DETR	GitHub	Efficient end-to-end detector	https://github.com/lyuwenyu/RT-DETR
70	A	RF-DETR	GitHub	Modern real-time DETR family reference	https://github.com/roboflow/rf-detr
71	B	FCOS	GitHub	Anchor-free detector baseline	https://github.com/tianzhi0549/FCOS
72	B	EfficientDet	GitHub	BiFPN detector reference	https://github.com/google/automl/tree/master/efficientdet
73	B	YOLOv5	GitHub	S2ADet-compatible historical base	https://github.com/ultralytics/yolov5
74	B	timm	GitHub	Pretrained backbone library	https://github.com/huggingface/pytorch-image-models
75	A	SAHI	GitHub	Sliced small-object inference	https://github.com/obss/sahi
76	B	Weighted Boxes Fusion	GitHub	Same-model TTA box merging only if legal	https://github.com/ZFTurbo/Weighted-Boxes-Fusion
77	B	Soft-NMS paper	Paper	Postprocessing option	https://arxiv.org/abs/1704.04503
78	B	VisDrone dataset	Dataset	Small-object detection methodology reference	https://github.com/VisDrone/VisDrone-Dataset
79	B	xView dataset	Dataset	Dense/small aerial detection reference	https://challenge.xviewdataset.org/
80	B	FiftyOne	GitHub	Detection error analysis/visualization	https://github.com/voxel51/fiftyone
81	A	Albumentations	GitHub	Geometry augmentation pipeline	https://github.com/albumentations-team/albumentations
82	B	MixUp paper	Paper	Controlled mixing reference	https://arxiv.org/abs/1710.09412
83	B	CutMix paper	Paper	Region mixing reference	https://arxiv.org/abs/1905.04899
84	B	Copy-Paste paper	Paper	Object copy-paste reference	https://arxiv.org/abs/2012.07177
85	B	Focal Loss paper	Paper	Class imbalance / hard examples	https://arxiv.org/abs/1708.02002
86	B	Generalized Focal Loss	GitHub	Quality-aware detection losses	https://github.com/implus/GFocal
87	B	VarifocalNet	GitHub	IoU-aware classification score	https://github.com/hyz-xmaster/VarifocalNet
88	B	SIoU Loss paper	Paper	Box regression candidate	https://arxiv.org/abs/2205.12740
89	S	Spectral Python (SPy)	GitHub	HSI I/O, visualization, spectral analysis	https://github.com/spectralpython/spectral
90	A	Rasterio	GitHub	Raster I/O and transforms	https://github.com/rasterio/rasterio
91	A	TorchGeo	GitHub	Geospatial datasets/samplers/model utilities	https://github.com/microsoft/torchgeo
92	A	TerraTorch	GitHub	EO foundation-model fine-tuning framework	https://github.com/IBM/terratorch
93	B	scikit-image	GitHub	Image transforms/SSIM/processing	https://github.com/scikit-image/scikit-image

					image
94	B	SciPy signal	Docs	Savitzky-Golay spectral smoothing/derivatives	https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.savgol_filter.html
95	B	scikit-learn PCA	Docs	Train-only PCA / IncrementalPCA	https://scikit-learn.org/stable/modules/decomposition.html#pca
96	B	OpenCV	GitHub	Geometry transforms, visualization	https://github.com/opencv/opencv
97	B	h5py	GitHub	HDFS cube I/O if applicable	https://github.com/h5py/h5py
98	B	xarray	GitHub	Labeled N-D spectral data analysis	https://github.com/pydata/xarray
99	A	Weights & Biases	Docs	Experiment/metric/artifact tracking	https://docs.wandb.ai/
100	B	MLflow	GitHub	Run and artifact tracking	https://github.com/mlflow/mlflow
101	B	Optuna	GitHub	Constrained hyperparameter tuning	https://github.com/optuna/optuna
102	B	Hydra	GitHub	Config-managed experiments	https://github.com/facebookresearch/hydra
103	B	PyTorch Lightning	GitHub	Training-loop reproducibility	https://github.com/Lightning-AI/pytorch-lightning
104	A	pycocotools	GitHub	COCO-style AP evaluation reference	https://github.com/cocodataset/cocoapi
105	B	COCO evaluation metrics	Website	AP50-95 conventions	https://cocodataset.org/#detection-eval
106	B	Temperature Scaling	Paper	Confidence calibration reference	https://arxiv.org/abs/1706.04599
107	B	Deep Ensembles paper	Paper	Uncertainty concept only; final multi-model ensemble not allowed	https://arxiv.org/abs/1612.01474
108	B	Mean Teacher	Paper	Teacher-student semi-supervised concept	https://arxiv.org/abs/1703.01780
109	B	Unbiased Teacher	GitHub	Semi-supervised object detection pattern	https://github.com/facebookresearch/unbiased-teacher
110	B	Soft Teacher	GitHub	Pseudo-label object detection pattern	https://github.com/microsoft/SoftTeacher
111	B	Dense Teacher	GitHub	Dense pseudo-label learning reference	https://github.com/Megvii-BaseDetection/DenseTeacher
112	B	MMDetection semi-supervised ecosystem	GitHub	Teacher-student implementation reference	https://github.com/open-mmlab/mmdetection
113	B	Segment Anything	GitHub	ROI foreground-mask research	https://github.com/facebookresearch/segment-anything
114	B	GroundingDINO	GitHub	Open-vocabulary detector research only	https://github.com/IDEA-Research/GroundingDINO
115	B	ConvNeXt	GitHub	Strong convolutional backbone	https://github.com/facebookresearch/ConvNeXt
116	B	Swin Transformer V2	Paper	High-resolution hierarchical transformer	https://arxiv.org/abs/2111.09883
117	B	ConvNeXt V2	Paper	Masked pretraining conv backbone	https://arxiv.org/abs/2301.00808
118	B	EVA	GitHub	Large pretrained visual backbone research	https://github.com/baaivision/EVA
119	B	DINOv2 paper	arXiv	Robust visual representations	https://arxiv.org/abs/2304.07193
120	C	Papers With Code - Object Detection	Benchmark	Broad detector benchmark lookup	https://paperswithcode.com/task/object-detection
121	C	Papers With Code - Hyperspectral Image Classification	Benchmark	HSI architecture lookup	https://paperswithcode.com/task/hyperspectral-image-classification
122	C	Hugging Face model search: hyperspectral	HF Search	Discover newly released HSI weights	https://huggingface.co/models?search=hyperspectral
123	C	GitHub search: hyperspectral object detection	GitHub Search	Discover new implementations	https://github.com/search?q=hyperspectral+object+detection&type=repositories

11. Claude Code Master Prompt

You are my senior Kaggle Grandmaster-level hyperspectral object-detection engineer. Work on the Kaggle Hyperspectral Object Detection Challenge 2026 using the attached HOD26 resource/experiment report as the research basis.

OBJECTIVE

Build the strongest robust SINGLE-CHECKPOINT solution we can justify with leakage-safe local validation. Do not chase public LB blindly and do not fabricate scores.

HARD RULES

1. Re-check the live competition rules and organizer clarifications before coding.
2. Final submission must come from ONE trained model / ONE checkpoint.
3. Same-model TTA/multi-scale is allowed only if the current organizer wording still allows it.
4. Public pretrained weights may be used only when legal; log model, source, URL and license.
5. Do not use external labelled data for training unless the organizer explicitly allows it.
6. Unlabeled test pseudo-label self-training may be used only if the current rule still permits it; log teacher, thresholds, filters and iterations.
7. Ranking images are inference-only. Never train, pseudo-label, self-supervise, distill, fine-tune, recalibrate BatchNorm, entropy-minimize or otherwise adapt model/statistics on ranking images.
8. Never create a fold/model/architecture ensemble for the final submission.

PHASE 1 - DATA FORENSICS

Create a reproducible notebook/script that reports: file counts, cube shape, dtype, 16-band ordering/wavelength metadata, per-band min/max/mean/std/percentiles, correlations, PCA explained variance, bbox counts, class counts, bbox size/aspect ratio, objects/image, edge boxes, duplicate/near-duplicate clusters, and train-vs-test unlabeled distribution statistics. Save plots and CSV summaries. Do not infer hidden labels.

PHASE 2 - VALIDATION

Build a primary StratifiedGroupKFold or GroupKFold using the safest scene/acquisition/duplicate grouping available. If metadata does not provide groups, derive train-only duplicate/scene clusters and document the method. Also build one stress split. Metrics: mAP50-95, AP50, AP75, per-class AP, AP by object size, pairwise real/fake confusion, fold std and seed std.

PHASE 3 - BASELINE

Implement E0 clean baseline with deterministic preprocessing and one reproducible seed. Prefer YOLO11m or the strongest simple detector that fits available GPU. Keep all configs and predictions.

PHASE 4 - CONTROLLED EXPERIMENTS

Run in this order, changing ONE major variable at a time:

- E1 normalization;
- E2 selected-band pseudo-RGB vs train-only PCA;
- E3 learned 16->3 1x1 spectral adapter;
- E4 native 16-channel stem;
- E6 adjacent spectral-derivative branch;
- E5 S2ADet-inspired spectral/spatial dual stream;
- E11 resolution/P2/tile study if bbox analysis shows small objects;
- E8 HyperVision first, then HyperSL/HyperSIGMA only if resources allow;
- E9 wavelength-aware HyperFree/DOFA-style embedding using actual metadata wavelength centers;
- E10 SpecDETR/MMDetection benchmark if compute permits;
- E13 conservative spectral augmentation;
- E12 localization/loss tuning;
- E14 imbalance handling;
- E15 legal pseudo-label self-training;
- E16 spectral-consistency pseudo-label filtering;
- E17 OOF confidence/NMS calibration;

E18 same-checkpoint TTA;
E19 3-seed robustness test for finalists.

EXPERIMENT GATE

For each run save: experiment_id, parent_id, hypothesis, exact diff, fold, seed, model, input representation, pretrained source/license, image size, batch, optimizer, LR, epochs, augmentations, CV mAP50-95/AP50/AP75, per-class AP, fold std, stress-CV score, runtime, VRAM, checkpoint and notes.

A change is kept only if the gain is reproducible across the relevant validation tests. Public LB is supporting evidence only. If public LB rises while local/stress CV worsens, mark PUBLIC_LB_OVERFIT_RISK.

SPECTRAL-SPECIFIC REQUIREMENTS

- Preserve band order and wavelength metadata.
- Do not use RGB HSV augmentation on raw spectral bands.
- Test first derivative: $dB_i = B_{(i+1)} - B_i$.
- For real/fake pairs, report raw-spectrum and derivative separability and pairwise AP/confusion.
- For the best model, run channel ablation / spectral-attention analysis to verify that the model uses spectral information.

PSEUDO-LABEL REQUIREMENTS

Use only the allowed unlabeled test images. Start with high-confidence predictions from one teacher checkpoint. Compare confidence-only filtering against confidence + train-only spectral-prototype consistency (spectral angle/cosine). Keep pseudo loss weight conservative. Do not touch ranking images.

INFERENCE

Use exactly one final checkpoint. Tune confidence/NMS from OOF predictions. Test only same-checkpoint TTA transformations that improve OOF results. Generate the exact Kaggle CSV schema and validate coordinates/classes/confidences before submission.

IMPLEMENTATION OUTPUTS

Create:

- notebooks/00_data_forensics.ipynb
- notebooks/01_validation.ipynb
- src/data/
- src/models/spectral_adapter.py
- src/models/spectral_derivative.py
- src/models/s2a_fusion.py
- src/train.py
- src/validate.py
- src/infer.py
- src/pseudo_label.py
- src/make_submission.py
- configs/e00_baseline.yaml ... configs/e19_*.yaml
- experiments.csv
- resources_used.md
- RULE_COMPLIANCE.md
- README.md

IMPORTANT

Do not report an experiment as improved until it has actually run. Do not invent metrics. If a repository cannot be reproduced, mark it and move on. Preserve a clean baseline and implement changes incrementally so every gain can be attributed to one hypothesis.

Start with Phase 1 and Phase 2. Then implement E0. After E0 is validated, proceed through the experiment queue in order and maintain experiments.csv continuously.

12. Verification Notes

This report separates current competition rules from research inspiration. Competition facts and rule constraints were checked against the Kaggle competition overview and organizer discussion posts on 26 September 2026. Public repositories and model pages were used as technical references; any checkpoint introduced into a final solution must still have its exact source/license recorded and rechecked against the live rules.